Question # 10 of 10 (Start time: 09:41:07 PM, 04 July 2024)

The volume of the solid bounded by planes x=a and x=b with cross-sectional area A(x) perpendicular to the x-axis is

Select the correct option

Re

○	$V = \int_a^b A(x)dy$
○	$V = \int_a^b A(y)dy$
○	$V = \int_a^a A(y)dx$
●	$V = \int_a^b A(x)dx$

Click to Save Answer & Mo

Question # 9 of 10 (Start time: 09:40:21 PM, 04 July 2024)

The volume of a cylinder is the area of a cross section of the cylinder multiplied by the _____ of the cylinder.

Select the correct option

○	Diameter
○	Radius
○	Base
●	Height

Click to Save Answ

Question # 8 of 10 (Start time: 09:39:31 PM, 04 July 2024)

For a function $f(x)$ that is decreasing on the interval $[a, b]$, how would this affect the volume of the solid of revolution?

Select the correct option

Rel

- ○ No effect on volume.
- ○ Volume increases.
- ◉ Volume decreases.
- ○ Cannot be determined.

Click to Save Answer & Mov

Question # 7 of 10 (Start time: 09:38:13 PM, 04 July 2024) **Total Marks:** 1

$$If\ f\ continuous\ on\ [a,b]\ and\ F(x)=\int_a^x f(t)dt,\ then$$

Select the correct option

Reload Math Equations

○	none of these
○	$F^{/}(t)=f(x)\ on\ \ [a,b]$
○	$F^{/}(x)=f(t)\ on\ \ [a,b]$
○	$F^{/}(x)=f(x)\ on\ \ [a,b]$

Click to Save Answer & Move to Next Question

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MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 6 of 10 (Start time: 09:37:24 PM, 04 July 2024)

What is the purpose of using the washer method instead of the disk method?

Select the correct option

○	To simplify the integration process
○	To increase accuracy
●	To handle shapes with holes.
○	To reduce the number of slices

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Question # 5 of 10 (Start time: 09:36:31 PM, 04 July 2024)

The value of $\int_1^x y^2 dy =$ _____.

Select the correct option

- $\frac{y^3}{3} - \frac{1}{3}$
- $\frac{x^3}{3} - \frac{1}{3}$

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Question # 4 of 10 (Start time: 09:35:46 PM, 04 July 2024)

The expressions $x^2 + x, x^2 + x + 5, x^2 + x - 3$ have the same ..

Select the correct option

○	Anti-derivative
◉	Derivative

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MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 3 of 10 (Start time: 09:35:06 PM, 04 July 2024)

When using the technique of slicing to find the volume of an irregular solid, what is the basic idea?

Select the correct option

◉	Divide the solid into slices and sum their volumes.
○	Divide the solid into cylinders and sum their volumes.
○	Divide the solid into cubes and sum their volumes.
○	Measure the distance between slices.

Click to Sa

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 2 of 10 (Start time: 09:34:34 PM, 04 July 2024)

If the upper limit of Definite Integral is equal to its lower limit,then the value of Definite Integral will be ___

Select the correct option

○	Same
○	1
●	Zero
○	None of the above

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MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 1 of 10 (Start time: 09:33:48 PM, 04 July 2024)

If the radius of a sphere is doubled, how does the volume change?

Select the correct option

○	It becomes four times of the original volume.
◉	It becomes eight times of the original volume.
○	It becomes half of the original volume.
○	It remains the same of the original volume.